

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 04



By- Vishal Sir



Recap of Previous Lecture



Topic

Types of functional dependency ✖

Topic

Properties of functional dependency

Topics to be Covered



✓
Topic

Different types of keys in RDBMS

✓
Topic

Candidate key

✓
Topic

Super key

Note:- In a relational table no two tuples
Can be exactly same
→ Key is a very important concept in RDBMS

Key = A set of attributes that can uniquely identify
each tuple of a relation is called a key



Topic : Different types of keys

There are various types of keys in RDBMS

① Candidate Key

② Primary Key

③ Alternate Key

④ Super Key

⑤ Foreign Key

Will be
discussed in
ER diagram

- Minimal Set : - Minimal set is a set from which no element can be removed without destroying the associated property

From the set if any element can be removed without destroying its property, then it is not a minimal set.



Topic : Candidate key

i.e. a set of attributes from which no attribute can be removed without destroying its property of being a key

The minimal set of attributes that can uniquely identify each tuple in the relation is called a Candidate Key

(or)

The minimal set of attributes that can determine all the attributes of the relation is called a Candidate key

Every key in a relation need not be a minimal key. If key is minimal key then it can be called a Candidate key

If key is not minimal then we can not call it a Candidate key but it may still be a key.

Note:

If values of a set of attributes are guaranteed to be unique in a relation, then that set of attributes will always be a key of that relation { But may or may not be minimal }

eg. Let Relation "Student" define the information about students in a class.

Roll. No.	Sname

Roll. No. will be unique for every student of the class. ∴ Roll. No. will be a key of this Relⁿ

Sname may be same for two different students of the same class

eg: 1

Consider the following relation
Student (Sid, Sname, fee)
and functional dependencies that
exists in relation Student are

$\left. \begin{array}{l} \text{Sid} \rightarrow \text{Sname} \\ \text{Sid} \rightarrow \text{fee} \end{array} \right\}$

"Student"

Sid	Sname	fee
S ₁	A	200
S ₂	A	300
S ₃	B	250
S ₄	B	250
S ₅	C	300

W.r.t. given FDs we can observe that
"Sid" will always be unique in the tuples of
relation Student.
∴ "Sid" is a key of relation Student

eg: 2

Consider the following relation
Enroll (Sid, Cid, Inst-id)

and functional dependencies that
exists in relation Enroll is

$\text{Cid} \rightarrow \text{Inst-id}$

$\text{Sid, Cid} \rightarrow \text{Inst-id, Sid}$

$\text{Sid, Cid} \rightarrow \text{Inst-id, Sid, Cid}$

Enroll

Sid	Cid	Inst-id
S ₁	C ₁	101
S ₂	C ₁	101
S ₃	C ₂	103
S ₃	C ₃	102
S ₄	C ₄	103

Together the values of (Sid, Cid) will be
unique in all tuples of relation Enroll.
∴ (Sid, Cid) is a key of relation Enroll

i.e.

If A is a
key of the
relation

then

Any superset of
A will also be
a key of the
given relation

If values w.r.t.
attribute set 'A'
are unique in
all the tuples of
relation,

then values w.r.t.
attributes of any
superset of A will
also be unique in all
the tuples of relation

A	B	...
1		
2		
3		
4		
5		

Eg. 1 Consider the following relation
 Student (Sid, Sname, fee)
 and functional dependencies that
 exists in relation Student are

$\left. \begin{array}{l} \text{Sid} \rightarrow \text{Sname} \\ \text{Sid} \rightarrow \text{fee} \end{array} \right\}$

"Student"

Sid	Sname	fee
S ₁	A	200
S ₂	A	300
S ₃	B	250
S ₄	B	250
S ₅	C	300

W.r.t. given FDs we can observe that
 "Sid" will always be unique in the tuples of
 relation "Student".

- ∴ "Sid" is a key of relation Student
- (Sid, Sname) is a Superset of "Sid"
- ∴ (Sid, Sname) is also a key of relⁿ

Eg. 2 Consider the following relation
 Enroll (Sid, Cid, Inst-id)
 and functional dependencies that
 exists in relation Enroll is

$\text{Cid} \rightarrow \text{Inst-id}$

$\text{Sid, Cid} \rightarrow \text{Inst-id, Sid}$

$\text{Sid, Cid} \rightarrow \text{Inst-id, Sid, Cid}$

Enroll

Sid	Cid	Inst-id
S ₁	C ₁	101
S ₂	C ₁	101
S ₃	C ₂	103
S ₃	C ₃	102
S ₄	C ₄	103

Together the values of (Sid, Cid) will be
 unique in all tuples of relation Enroll.

- ∴ (Sid, Cid) is a key of relation Enroll
- (Sid, Cid, Inst-id) is a Superset of (Sid, Cid)
- ∴ (Sid, Cid, Inst-id) is also a key of relⁿ Enroll.

eg. 1 Consider the following relation
Student (Sid, Sname, fee)
and functional dependencies that
exists in relation Student are

$\left. \begin{array}{l} \text{Sid} \rightarrow \text{Sname} \\ \text{Sid} \rightarrow \text{fee} \end{array} \right\}$

"Student"		
Sid	Sname	fee
S ₁	A	200
S ₂	A	300
S ₃	B	250
S ₄	B	250
S ₅	C	300

W.r.t. given FDs we can observe that
"Sid" will always be unique in the tuples of
relation "Student".

∴ "Sid" is a Key of relation Student
(Sid, Sname) is a Superset of "Sid"
∴ (Sid, Sname) is also a key of relⁿ

In eg. 1

(Sid, Sname) is a key of relation "Student",
but it is not minimal, because
even if we remove attribute "Sname"
from the set, then "Sid" alone is
a key of the relation.

∴ (Sid, Sname) is not a candidate
key of the relation.

In eg. 1 "Sid" is also a key of the
relation "Student", and it is a
minimal key.

∴ "Sid" is a Candidate key of
relation Student.

Eg. 2

Consider the following relation
Enroll (Sid, Cid, Inst-id)

and functional dependencies that
exists in relation Enroll is

$Cid \rightarrow Inst-id$

$Sid, Cid \rightarrow Inst-id, Sid$

$Sid, Cid \rightarrow Inst-id, Sid, Cid$

Enroll

Sid	Cid	Inst-id
S ₁	C ₁	101
S ₂	C ₁	101
S ₃	C ₂	103
S ₃	C ₃	102
S ₄	C ₄	103

Together the values of (Sid, Cid) will be
Unique in all tuples of relation Enroll.

∴ (Sid, Cid) is a key of relation Enroll

(Sid, Cid, Inst-id) is a Superset of (Sid, Cid)

∴ (Sid, Cid, Inst-id) is also a key of relⁿ Enroll.

In eg. 2:

(Sid, Cid) is a key of relation Enroll.

* If we remove "Sid" from the set
then "Cid" alone can not become
a key of the relation "Enroll"

and * If we remove "Cid" from the
set then "Sid" alone can not become
a key of the relation "Enroll"

i.e. from the set (Sid, Cid) nothing can
be removed without destroying its
Property of being a key.

∴ (Sid, Cid) is a minimal set w.r.t. key

And Hence, (Sid, Cid) is a Candidate key of Relⁿ



Topic : Candidate key

- ① A key with a single attribute is always minimal.
∴ A key formed of a single attribute is always a Candidate Key.
- ② If candidate key is formed of a single attribute, then it is called a Simple candidate key.
- ③ If a candidate key is formed of two or more attributes then it is called "Composite" Candidate key.
- ④ A relation may have more than one Candidate Key.



Topic : Candidate key

⑤ Attributes belonging to any of the Candidate Key are called prime attributes. Otherwise non-prime attributes.

eg: let $R(A, B, C, D, E, F, G)$ is a relation and (AB) and (BCD) are the only two candidate keys of relation R , then

Prime attributes = $\{A, B, C, D\}$

Non-prime attributes = $\{E, F, G\}$

⑥ Attributes belonging to Candidate Key may be allowed to take NULL values.



Topic : Primary key

- { There may be multiple candidate keys in a relation,
at most one of those Candidate Keys can be
defined as a "primary key" of the relation
- In a relation there can be at most one primary key.
0 or 1
- Attributes of primary key are not allowed to take
"NULL" values.



Topic : Alternate key

{ Also known as secondary key } PW

All candidate keys except primary key are also known as alternate key.

All candidate keys

Primary key (at most 1 C.K.)	Alternate Keys
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All these keys
are "minimal keys"





Topic : Super key



{ it need not be minimal }
{ May or may not be minimal }

The set of attributes that can uniquely identify each tuple of the relation is called a Super key { May or may not be minimal }

Every Candidate Key is also a Super key,
but Every Super key need not be a Candidate Key

Every Super key is a Superset of some Candidate Key
is: one or more

Note:

At least one subset of a set of attributes of Superkey is a Candidate Key.

Note:

Super Key = All attributes of at least one Candidate Key + 0 or more attributes out of remaining attributes of the relation

eg. Consider the relation Student (Sid, Sname, Branch)
if "Sid" is the only candidate key of relation Student.
then find all Superkeys of relation Student.

• "Sid" is the only C.K.

∴ "Sid" must be present in all Superkey

• Any Superset of "Sid" will be a Superkey of relation Student.

Supersets of
 $\{ \text{Sid} \}$
are $\Rightarrow$

$$\{ \text{Sid} \} = \{ \text{Sid} \} \cup \{ \}$$

$$\{ \text{Sid}, \text{Sname} \} = \{ \text{Sid} \} \cup \{ \text{Sname} \}$$

$$\{ \text{Sid}, \text{Branch} \} = \{ \text{Sid} \} \cup \{ \text{Branch} \}$$

$$\{ \text{Sid}, \text{Sname}, \text{Branch} \} = \{ \text{Sid} \} \cup \{ \text{Sname}, \text{Branch} \}$$

Subsets of remaining
Attributes { Sname, Branch }

"4" subsets are
possible.
∴ "4" Superkeys

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R .
(i) when (A_1) is the only candidate key of $rel^h R$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R .
(ii) when $(A_1 A_2)$ together is the only candidate key of relⁿ R

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R .
(iii) When (A_1) and (A_2) are the only two candidate keys of $rel^h R$

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R.
(iv) When (A_1A_2) & (A_2A_3) are the only two candidate keys of relⁿ R

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R .
(V) When (A_1A_2) & (A_3A_4) are the only two candidate keys of relⁿ R

H.W. Q: Consider a relational schema $R(A_1, A_2, A_3, \dots, A_N)$
Find the total number of superkeys possible in relation R .
(vi) When $(A_1), (A_2)$ & (A_3) are the three candidate keys of relⁿ R



2 mins Summary



Topic

Different types of keys in RDBMS ✓

Topic

Candidate key ✓

Topic

Super key ✓

THANK - YOU